1.2.4 Application of matrices to Geometry

There is a special type of non-singular matrices which are widely used in applications of matrices to geometry. For simplicity, we consider two-dimensional analytical geometry.

Let O be the origin, and $x'Ox$ and $y'Oy$ be the x-axis and y-axis. Let P be a point in the plane whose coordinates are (x, y) with respect to the coordinate system. Suppose that we rotate the x-axis and y-axis about the origin, through an angle θ as shown in the figure. Let $X'OX$ and $Y'OY$ be the new X-axis and new Y-axis. Let (X,Y) be the new set of coordinates of P with respect to the new coordinate system. Referring to Fig.1.1, we get

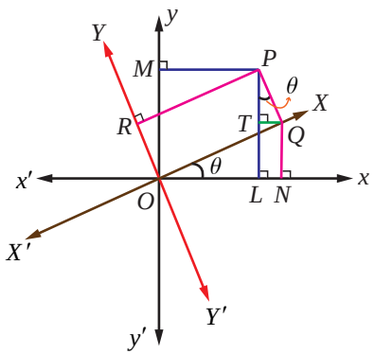


Fig.1.1

$$x = OL = ON - LN = X\cos\theta - QT = X\cos\theta - Y\sin\theta\,,$$

$$y = PL = PT + TL = QN + PT = X\sin\theta + Y\cos\theta\,.$$

These equations provide transformation of one coordinate system into another coordinate system. The above two equations can be written in the matrix form

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} X \\ Y \end{bmatrix}.$$

Let $W = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. Then $\begin{bmatrix} x \\ y \end{bmatrix} = W\begin{bmatrix} X \\ Y \end{bmatrix}$ and $|W| = \cos^2\theta + \sin^2\theta = 1$.

So, W has inverse and $W^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$. We note that $W^{-1} = W^T$. Then, we get the inverse transformation by the equation

$$\begin{bmatrix} X \\ Y \end{bmatrix} = W^{-1}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}.$$

Hence, we get the transformation $X = x\cos\theta - y\sin\theta$, $Y = x\sin\theta + y\cos\theta$.

This transformation is used in **Computer Graphics** and determined by the matrix $W = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. We note that the matrix W satisfies a special property $W^{-1} = W^T$; that is, $WW^T = W^TW = I$.

Definition 1.3

A square matrix A is called **orthogonal** if $AA^T = A^TA = I$.

Note

A is **orthogonal** if and only if A is non-singular and $A^{-1} = A^T$.

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